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Enhancing Reservoir Management with a Business Intelligence Tool for Permanent Downhole Gauge Monitoring

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Abstract

Permanent Downhole Gauges (PDHGs) are essential tools for reservoir and well management, providing continuous pressure and temperature data critical for optimizing performance and ensuring well integrity. Despite technological advancements, PDHG reliability remains a challenge due to harsh downhole conditions and telemetry limitations, particularly offshore. This paper presents the development of a Business Intelligence (BI) dashboard designed to enhance PDHG status monitoring and management across operated affiliates. By integrating real-time data collection, automated error detection, and survival analysis using the Kaplan-Meier probabilistic method, the tool enables proactive gauge replacement planning, improved gauge procurement decisions, and significant operational efficiencies in the long term. Case studies demonstrate the value of accurate downhole pressure and temperature data in reducing production drawdown uncertainty and in optimizing reservoir performance, highlighting the potential to transform PDHG reliability management with proper treatment of already-available data.

Introduction

In modern reservoir management, real-time data access is vital for optimizing hydrocarbon recovery and operational efficiency. PDHGs play a central role by continuously delivering high-resolution pressure and temperature data from the wellbore. Since their inception in the late 1980s, PDHGs have evolved from analog gauges to advanced digital systems with bidirectional communication and integration capabilities. Key innovations include Frequency Division Multiple Access-based telemetry for reliable data transmission and durable hardware enhancements (corrosion-resistant alloys, fully welded assemblies, and advanced transducers), extending system lifespans beyond 20 years (SLB, n.d., 2025).

PDHGs support applications such as pressure transient analysis, production optimization, and reservoir model calibration (Al-Khelaiwi and Davies, 2005). Their use has reduced costly well interventions and improved subsurface data granularity. Field deployments available in the literature—like those in the Gullfaks and Veslefrikk fields and offshore Nigeria—demonstrate their impact on production and reservoir characterization (Davies et al., 2005; Al-Khelaiwi and Davies, 2005, 2006). Nevertheless these benefits, reliability remains a concern. While survival rates improved in the early days, failures persist due to gauge

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and cable integrity issues, harsh downhole conditions, and telemetry incompatibilities with ESPs (Van Gisbergen and Vandeweijer, 2001; Mohammed et al., 2024). Emerging solutions include fiber optics and predictive maintenance strategies, as well as retrofit technologies.

This paper examines how integrating Business Intelligence (BI) tools with PDHG monitoring can enhance reservoir management by improving data accessibility, reliability analysis, and straight decision-making workflows.

Reliability Challenges of Permanent Downhole Gauges (PDHGs)

Reliability is a key concern in PDHG deployment due to the high cost and complexity of downhole interventions. Since their introduction, the reliability of PDHG systems has been a persistent concern, with early installations exhibiting 5-year survival rates as low as 40%. By the early 1990s, improvements in design and installation practices increased survival rates to 75%, but further progress plateaued, with failures predominantly attributed to electronics and cables (Van Gisbergen & Vandeweijer, 2001; Frota & Destro, 2006). The most relevant reliability issues are related to the following:

- Harsh Downhole Conditions: PDHGs are exposed to high pressure, high temperature (HPHT), corrosive fluids, and mechanical vibrations, all of which accelerate wear and can lead to gauge drift, seal failure, or electronic degradation (Van Gisbergen & Vandeweijer, 2001; Veneruso et al., 2000).
- Cable and Connector Failures: Mechanical damage during installation, thermal cycling, and chemical exposure can compromise insulation and signal integrity, with connectors and cables remaining the most common sources of early failures (Van Gisbergen & Vandeweijer, 2001; Veneruso et al., 2000).
- Limited Accessibility for Maintenance: Failures often require costly workovers or sidetracks, making reliability a top concern for operators, especially in offshore and subsea environments (Frota & Destro, 2006). In most cases, failed PDHGs are never replaced, and memory gauge campaigns are used to retrieve pressure data, at a high cost and associated risk.
- Telemetry and Data Transmission Issues: Signal attenuation, electromagnetic interference, and telemetry faults can disrupt data flow, particularly in long horizontal wells or multi-zone completions (Mohammed et al., 2024; Silva Junior et al., 2012).
- Compatibility with Artificial Lift Systems: PDHGs installed near ESPs or gas lift valves are exposed to high vibration and electrical noise, which can shorten lifespan unless fiber optic alternatives are used (Mohammed et al., 2024; Silva Junior et al., 2012).

To address these challenges, the industry has adopted structured reliability assurance processes, including Failure Mode and Effects Analysis (FMEA), accelerated life testing, and survival analysis of field data (Veneruso et al., 2000; Veneruso, Kohli & Webster, 2003). These methods have enabled the identification and mitigation of failure modes, particularly those associated with connectors and deployment procedures. For example, improvements in connector design and installation practices have reduced gauge failure rates and increased the 5-year reliability of PDHG systems, especially at temperatures below 100°C (Veneruso et al., 2000; Frota & Destro, 2006).

Survival analysis and Weibull modeling (Martz, 2004) have shown that failure rates are not constant over time, with early failures often linked to installation and manufacturing defects, while later failures are associated with wear-out mechanisms (Veneruso et al., 2000; Van Gisbergen & Vandeweijer, 2001). The adoption of more robust qualification testing and field data collection has been critical for continuous improvement (Veneruso et al., 2000; Frota & Destro, 2006).

Recent advances emphasize fault diagnosis and predictive maintenance to reduce non-productive time (NPT). Techniques such as condition monitoring and the use of large reliability databases are increasingly used to anticipate failures and schedule timely interventions (Sparke, 2015; Silva Junior et al., 2012). Fiber optic systems have emerged as a more reliable technological alternative in high-vibration environments,

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such as those involving Electric Submersible Pumps (ESPs), offering enhanced durability and data integrity, and potentially doubling operational lifespan (Silva Junior et al., 2012; Parker, 2014). In addition, the integration of distributed acoustic sensing (DAS) and distributed temperature sensing (DTS) technologies further extends the capabilities of downhole monitoring, enabling real-time surveillance over the entire wellbore and providing valuable data for both production optimization and equipment health monitoring (Parker, 2014; Silva Junior et al., 2012).

Despite these advances, achieving the industry target of 90% 5-year survival probability remains challenging, particularly in HPHT and subsea applications. Continued collaboration between operators, service companies, and equipment manufacturers is essential to drive further improvements in reliability, standardization of protocols, and the development of robust qualification and testing standards (Frota & Destro, 2006; Veneruso et al., 2000).

Inefficiencies of the Existing PDHG Performance and Reliability Management System from the Operator's Perspective

The previous system used for managing operated PDHGs suffered several challenges:

- Lack of an Exhaustive Vision on Gauge Status: The absence of a comprehensive overview of the status of PDHGs weighed down on effective technology selection, procurement, and preventive planning.
- Time-Consuming Gauge Status Data Collection: The process of collecting gauge data, based on annual surveys, required a great deal of labor-intensive work and could take months, leading to inefficiencies and delays. Annual surveys were conducted regularly by a central Reservoir Management team or on a need-basis by asset or country but require a large amount of data gathering and processing effort from reservoir and completion engineers. In addition, updates were inconsistent and unmethodical.
- Systemic Errors in Gauge Data Collection: Inevitable human-related errors in survey data collection, whether from service companies or internal sources, can lead to inaccurate assessments and decisions. This is particularly common with respect to gauge types, technology maturity, and reservoir conditions associated with a particular installation.
- Procurement Based on Price Alone: The lack of communication between project/completion engineers and reservoir engineers results in gauge procurement decisions being made almost solely on a price basis, as part of larger completion packages. Only minimum technical requirements are used to filter technologies, but we have constated that it is not enough, and that the long-term benefits should also be considered. The choice of the appropriate technology for the right bottomhole environment will eventually be translated into additional barrels, as shown in case studies at the end of this paper.

A process flow summarizing the previous methodology used for gauge status surveys, together with its main inefficiencies, is shown in Figure 1. An average of 4 months is needed to gather, QC, process and display data for one country affiliate at any given time by using an annual survey procedure. The process of obtaining QC'd data to update the gauge database was inefficient, particularly with respect to the amount of communications needed to identify the actors and find the information. After these inefficiencies were identified, a Customer Survey was launched with two large West Africa affiliates by targeting the main users of gauge data (completion, reservoir and monitoring engineers, data managers, database specialists) to detect the main improvement needs. Figure 2 shows a CTQ (Critical to Quality) diagram of the expected requirements by these users.

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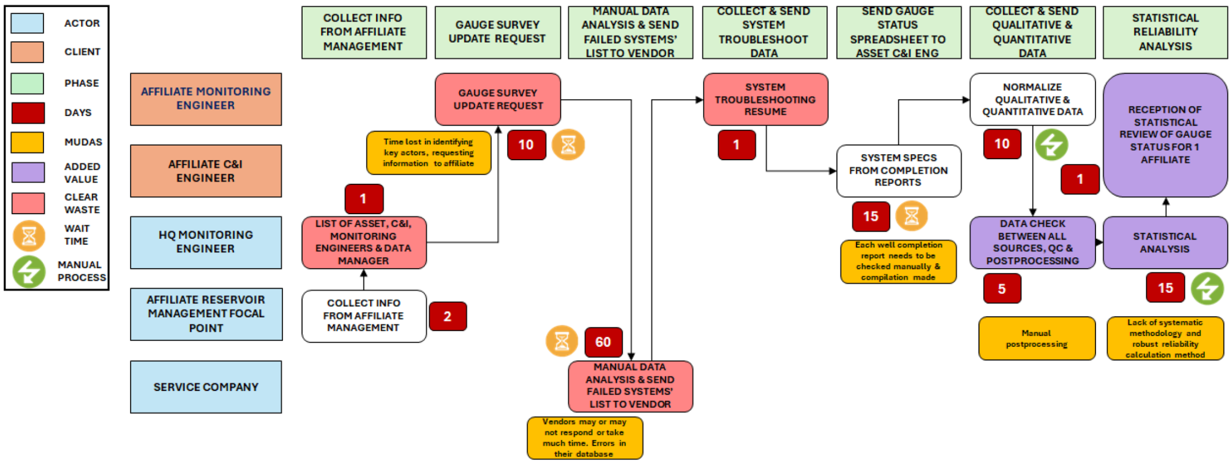

Figure 1—Process flow showing the inefficiencies in the methodology used previously to gather permanent downhole gauge information.

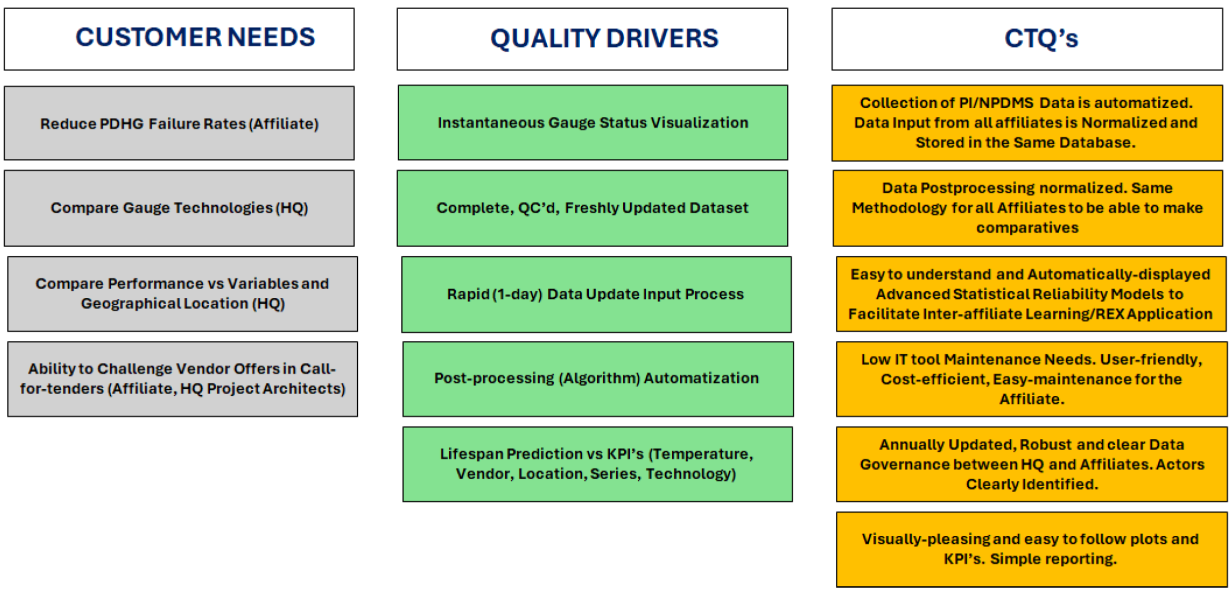

Figure 2—CTQ (Critical to Quality) diagram summarizing specific, measurable, and actionable performance requirements by the final users of the gauge data (completion and reservoir monitoring engineers, drilling procurement specialists).

New Methodology

We focused on developing a centralized PDHG dashboard designed to enhance monitoring and management of downhole gauges. This tool integrates automated survey data collection, real-time status updates, and a few simple algorithms for detecting and correcting errors in historical pressure and temperature data. By streamlining data acquisition and applying intelligent diagnostics, the dashboard reduces manual effort and improves data reliability. In addition to operational oversight, the dashboard supports procurement planning by enabling rapid comparisons between similar technologies. This ensures that purchasing decisions are aligned with technical requirements and long-term performance goals. The system contributes to improved efficiency by minimizing time spent on data handling and enhancing the accuracy of information used in decision-making. It also delivers cost savings through optimized gauge selection, which reduces intervention campaigns and production losses while supporting better reservoir management.

The dashboard provides a global view of gauge status and tracks installations by year, service company, and asset. It includes survivability and reliability metrics by technology, key performance indicators such as uptime, downtime, and lifespan, as well as gauge technical specifications. A built-in

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technology catalog helps identify the most suitable gauge for specific well conditions, supporting proactive planning and continuous data acquisition. Data sources include the in-house Reservoir Master Database (RMDB), historical Plant-Information Asset Framework (PI-AF®) data, the in-house New Production Data Management System (NPDMS) information, and Excel files containing vendor-provided technology information for each device. The process includes data preparation, integration, feature engineering, and data modeling. Survival analysis using the Kaplan-Meier method (Durczak et al., 2022, Sahadevan and Khan, 2024) is employed to estimate the reliability of PDHGs. In addition, ordinary statistical methods are used to predict life expectancy depending on a series of variables, particularly temperature.

PDHG Status Rules

Since the main purpose of the tool is to analyze PDHG performance and reliability via data analytics KPI's, it is key to be able to filter periods in which gauges send erroneous data. Various types of gauge data errors are common, and it is not efficient to evaluate each individual gauge manually; instead, data thresholds based on well-known reservoir engineering knowledge of each asset are used to filter out inconsistent data. Reservoir reference datum parameters are used to maintain consistency across fields. Gauge status is updated on a real-time basis, meaning that a gauge may change status several times (e.g. a gauge is off during an intervention but comes back online later). Figure 3 shows a cartoon overview of the criteria used to define gauge status and the main definitions. A Data Analytics eXpression (DAX) and Python code was used to apply these criteria in the BI tool, based on asset-specific historical benchmarks. Figure 4 shows an example of a typical issue encountered at a real data evaluation by the algorithm. Some common cases of data error filtering needs are: zero or negative values, values too deviated from the BHFP or datum conditions, periods of gauge inactivity, frozen data due to surface Data Acquisition System rebooting or maintenance, unmatching installation and first data transmission dates or wrong data acquired during well testing. Some of these errors are due to the fact that the algorithm averages real-time data to a daily mean. Thus, statistical checks are applied to automatically filter the erroneus data, providing a constant method to calculate KPI's across all fields.

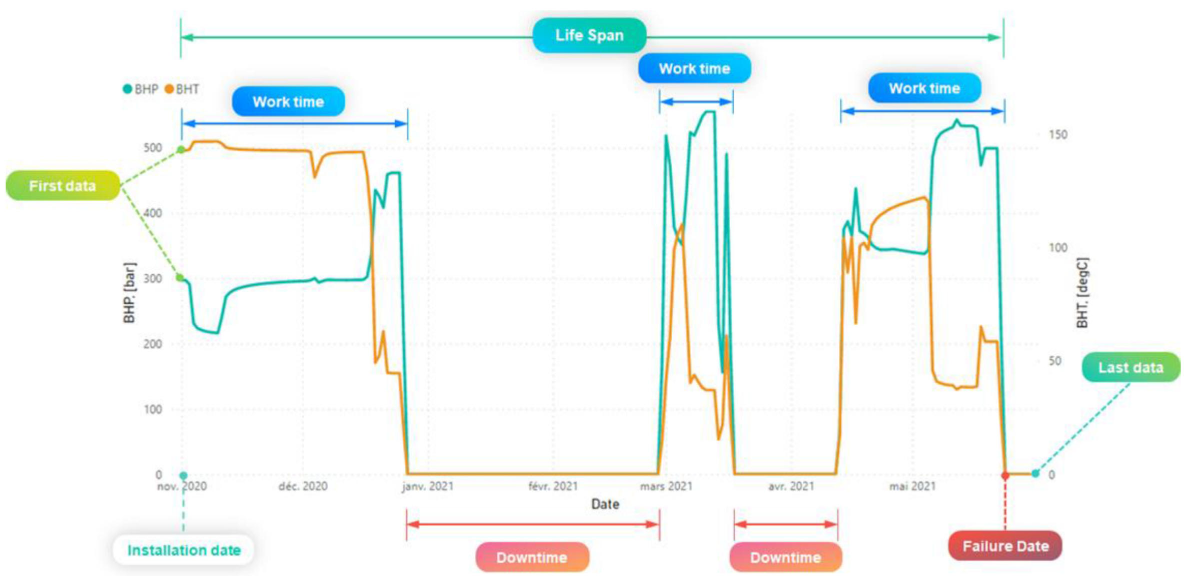

Figure 3—Criteria used to define the main gauge status at any given time.

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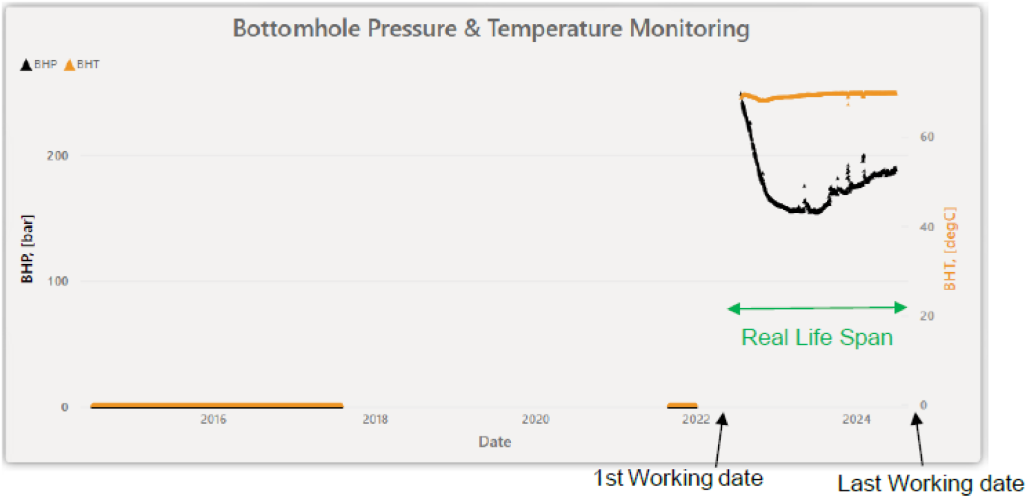

Figure 4—Real data evaluation example.

Survival Analysis

The main issue with permanent downhole gauges reliability estimates lies in that these instruments are installed downhole and thus there is no direct access to the gauge to evaluate the source of failure. Only limited troubleshooting can be done from the wellhead by using electronic transmission via the cable. Cause-root analysis, therefore, is very scarce and only occurs when a completion is pulled out of the hole for a full workover, an intervention which is very expensive, especially offshore.

Gauge reliability models, therefore, rely on statistical analysis, requiring large datasets to be meaningful. Combining an endpoint event and the time experienced by this event (in this case, a gauge failure) is used. In reliability engineering, this is used to estimate the probability of failure and the failure rate of devices, as well as to calculate the mean time between failures and the mean time to first failure. Figure 5 shows probabilistic Kaplan-Meier curves, displaying the estimated survival probability for 5 different service companies from more than 600 gauges installed in conventional offshore and deepwater wells since late 1990s.

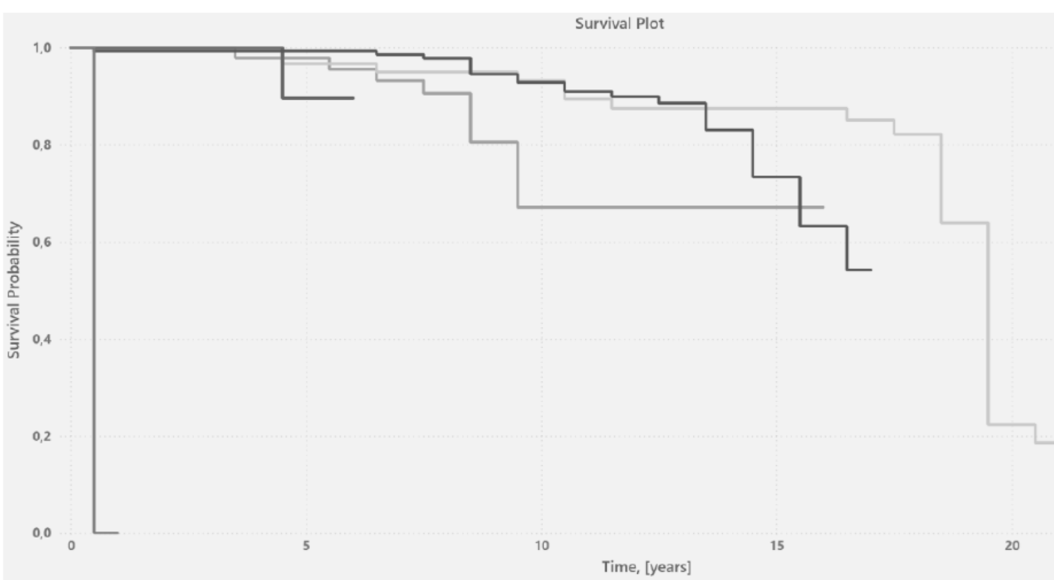

Figure 5—Kaplan-Meier curves displaying the estimated survival probability for five different service companies, from more than 600 gauges installed in conventional offshore and deepwater wells since 1980.

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Survival and Censoring Times

In order to estimate the probability of failure for a given family of gauges, survival and censoring times should first be calculated. For each individual gauge, a true survival time, (T), is assumed, as well as a true censoring time, (C). The survival time, also known as the failure time or event time, represents the time at which the event of interest occurs, such as when a device fails. In contrast, the censoring time is when censoring occurs, for example, when a device is disconnected intentionally or unintentionally or is broken. We can observe the random variable:

$$Y = \min(T, C)$$

In other words, if the event in question occurs before censoring (i.e., T < C), then we observe the true survival time T; however, if censoring occurs before the event (T > C) then we observe the censoring time (for example when a gauge works intermittently). We observe a status indicator:

$$\delta = \begin{cases} 1, & if\ T \le C \\ 0, & if\ T > C \end{cases}$$

Thus, δ = 1 if we observe the true survival time, and δ = 0 if we observe the censoring time.

Survival Function

The survival function, S(t), describes the probability of surviving past a specified time point (t), or more generally, the probability that the event of interest (the gauge failure) has not yet occurred by this time-point. The survival function is:

$$S(t) = 1 - F(t) = P(T > t) \quad \text{for} \quad t > 0$$

Where T denotes a continuous non-negative time until final failure.

Kaplan-Meier Method

The Kaplan-Meier curve shows the estimated survival function by plotting estimated survival probabilities against time (Figure 5). The estimated survival probability is constant between events, resulting in a step-function where each vertical drop indicates the occurrence of one or more events. The Kaplan-Meier model (Sahadevan and Khan, 2024) computes the survival function with the formula:

$$S(t) = \prod_{i:t_i \le t} \left(1 - \frac{d_i}{n_i}\right)$$

In medicine, where this method is mostly used for drug testing, censoring of patients is typically indicated by a vertical mark at the censoring time or other symbols like an asterisk. The same methodology is used here for gauge instruments.

Example of building a Kaplan-Meier curve and estimating its survival function

A basic mock data set of 7 downhole gauges with a follow-up of 6 years is shown in Table 1. All of them are "at risk" of having the failure event at time 0. Four gauges (A, D, F and G) have suffered an event (years 2, 3, 4, and 5, see Figure 6), whereas 3 gauges (B, C and E) have censored data: 2 gauges (C and E) are lost to follow-up (2 at year 3), and only 1 gauge (B) finishes the follow-up without a failure event (censored at year 6).

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Table 1—Gauge technologies A to G with their observed follow-up time and status.

Gauge	Follow-up (years)	Status
A	4	1
B	6	0
C	3	0
D	5	1
E	3	0
F	3	1
G	2	1

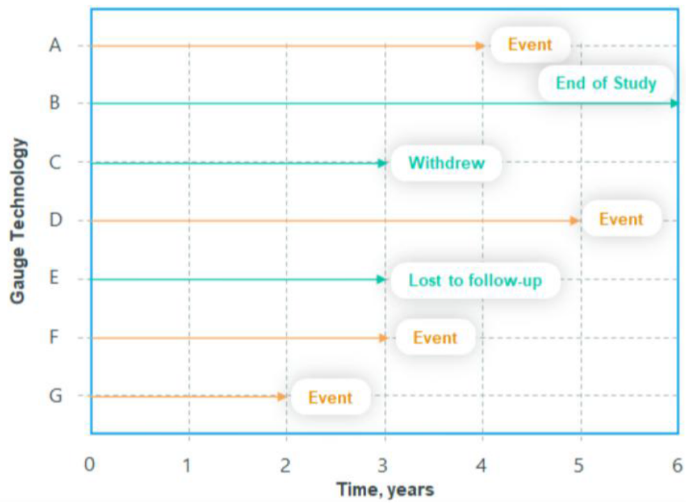

Figure 6—Graphical representation of downhole gauge events.

As seen in this example, the Kaplan-Meier survival estimate often takes the form of a step plot (Figure 7) rather than a smooth curve. This reflects that it is an empirical estimate of the survival experience of a 'gauge technology' at each time point. Importantly, censored observations do not reduce cumulative survival but rather adjust the number at risk for when the next event occurs.

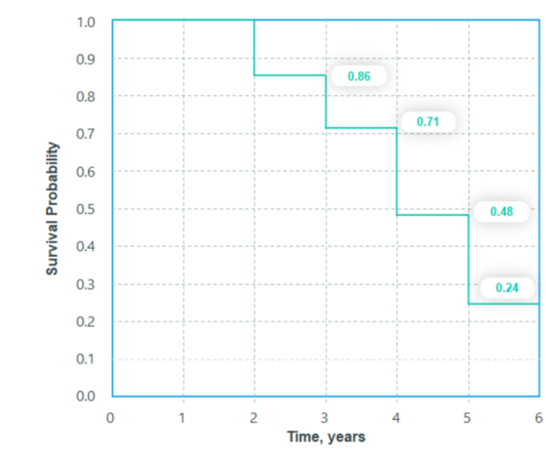

Figure 7—Kaplan-Meier survival probability vs time plot.

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The procedure to calculate the survival function is as follows. The cumulative probability of surviving to the end of an interval is estimated using conditional probabilities. To survive to the end of a period interval, a gauge must have survived all previous periods. This conditional probability is obtained by multiplying the sequence of the period-specific conditional survival probabilities. Table 2 and Figure 7 display an example of this procedure for the mock data.

Table 2—Estimated gauge survivor function based on conditional probabilities.

Survival time	Number of events	Number of censored	Number at risk	Probability of failure	Survival function
0	0	0	7	1-0/7 =1	1.000
1	0	0	7	1-0/7 =7/7	1*(7/7) = 1.000
2	1	0	7	1-1/7 = 6/7	1*(6/7) = 0.857
3	1	2	6	1-1/6 = 5/6	1*(6/7)*(5/6) = 0.714
4	1	0	3	1-1/3 = 2/3	1*(6/7)*(5/6)*(2/3) = 0.476
5	1	0	2	1-1/2 = 1/2	1*(6/7)*(5/6)*(2/3)*(1/2)= 0.238
6	0	1	1	--	--

Reliability Model

When the process explained above is fully automated, manual and systemic errors can be significantly reduced. The integration of survival analysis into a BI tool provides a unique perspective on PDHG reliability, enabling proactive planning to ensure bottomhole pressure measurements are not discontinued, and better procurement decisions. Figure 8 shows the results of a real survival analysis for the number of gauge systems installed in one of our assets. Each curve represents a specific gauge technology.

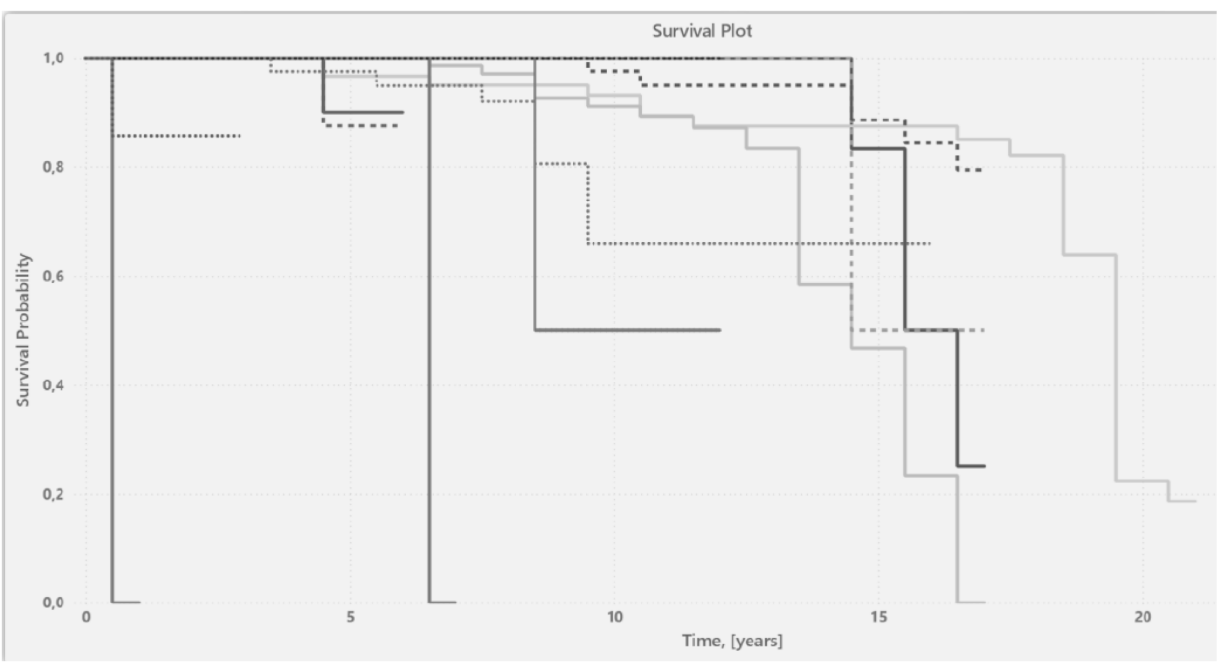

Figure 8—Example of a survival analysis. Each curve corresponds to the survival probability for an individual commercial gauge system. This track record plot is used to select the gauge technologies for a new installation based on robust data, complementing service company recommendations.

The addition of the automated data gathering and processing BI dashboard leads to a simplified workflow with significant cost and time savings with respect to the previous methodology. Figure 9 shows the

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new process flow showing the methodology used currently to gather permanent downhole gauge status information. Data gathering is fully-automatic and manual checks are only added when the system detects gauge status changes or new wells are added to the database.

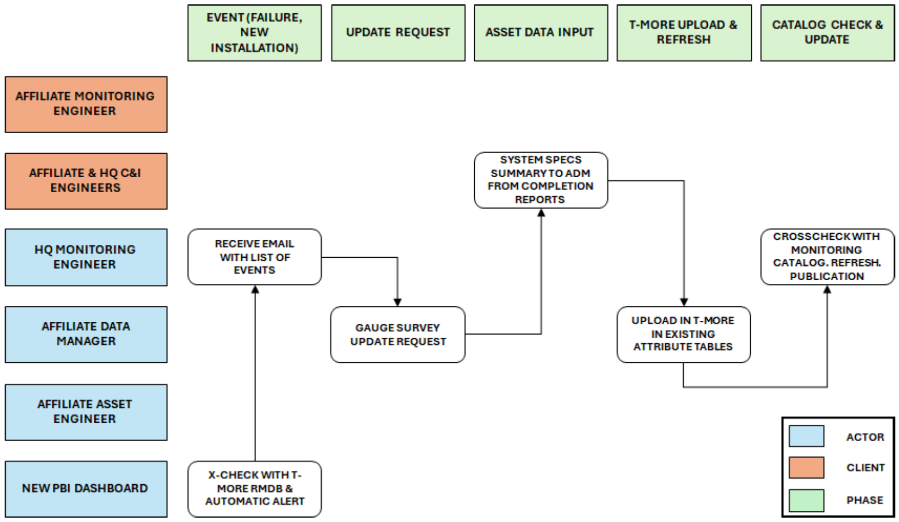

Figure 9—New process flow showing the methodology used currently to gather permanent downhole gauge information.

Other than the probabilistic approach, additional analysis can be done by calculating gauge performance KPIs, as shown in Figure 10. Individual gauge monitoring in real time (Figure 11) and analysis of the survival probability versus temperature and other parameters may also provide useful insights on gauge performance.

Figure 10—Example of KPI display for three different gauge systems from the same vendor.

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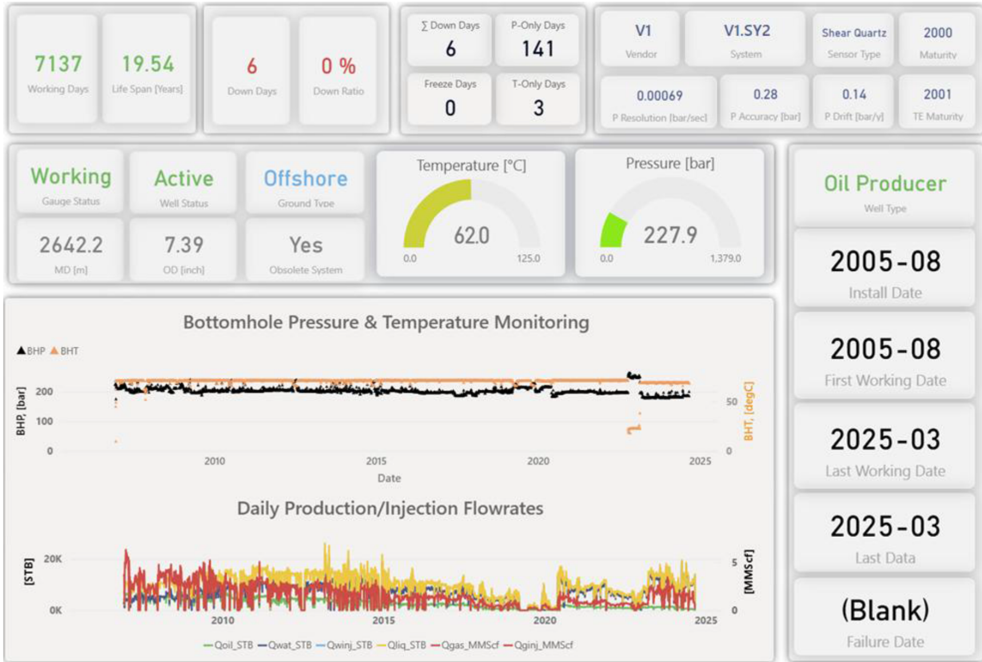

Figure 11—Display of the real-time monitoring parameters for an individual gauge installed in an offshore field.

Business Impact

In addition to improving data analysis efficiency and reducing engineers' time dedicated to this task, the implementation of this PDHG dashboard is expected to result in several key operational benefits in the long term:

- Improved Efficiency: Reduced time and effort in data collection and management (from 3-4 months to 1 week), and access to more reliable and accurate data for procurement decision-making and/or pressure acquisition campaign planning.
- Cost Savings: Optimized gauge choices leading to cost savings in Operating Expenditures (OPEX), including less intervention campaigns, less production shortfalls and better well and reservoir management, including:
  - Reduced long-term OPEX intervention costs (particularly a decrease in the number of memory gauge PBUs) in new installations, estimated at 2-3 MM$ per offshore well.
  - Reduced emissions per well by decreasing the need for annual well interventions (boat trips to wellhead)
- Production Optimization (increased NPV):
  - Increase in production potential per well by reducing BHFP uncertainty. This is particularly critical when trying to optimize drawdown on a well equipped with sand control. In a typical plateau offshore well, optimization can add up to 1-2MMbbl of cumulated production.
  - Increase in field-wide production potential by reducing full-field reservoir model BHFP uncertainty. Estimates are that the impact may be up to 1-3% of the overall field production.

A Value Of Information (VOI) case study from a deepwater field is presented in Figure 12, illustrating uncertainties from the early loss of a downhole gauge and the potential cumulative production if a PDHG were still in place. Well DW-1, completed in the most productive system of the field, operates under substantial uncertainty due to the lack of a functional PDHG since 2003. BHFP is estimated indirectly by using wellhead flowing pressure and riser test (well performance) models, which are sensitive to GOR

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and BSW variations. These introduce a ±7 bar margin in drawdown estimation, risking underestimation of productivity or overestimation of well integrity threats—especially sand production, given the well's drawdown of 48 bars and liquid rate of 19 kbbl/d.

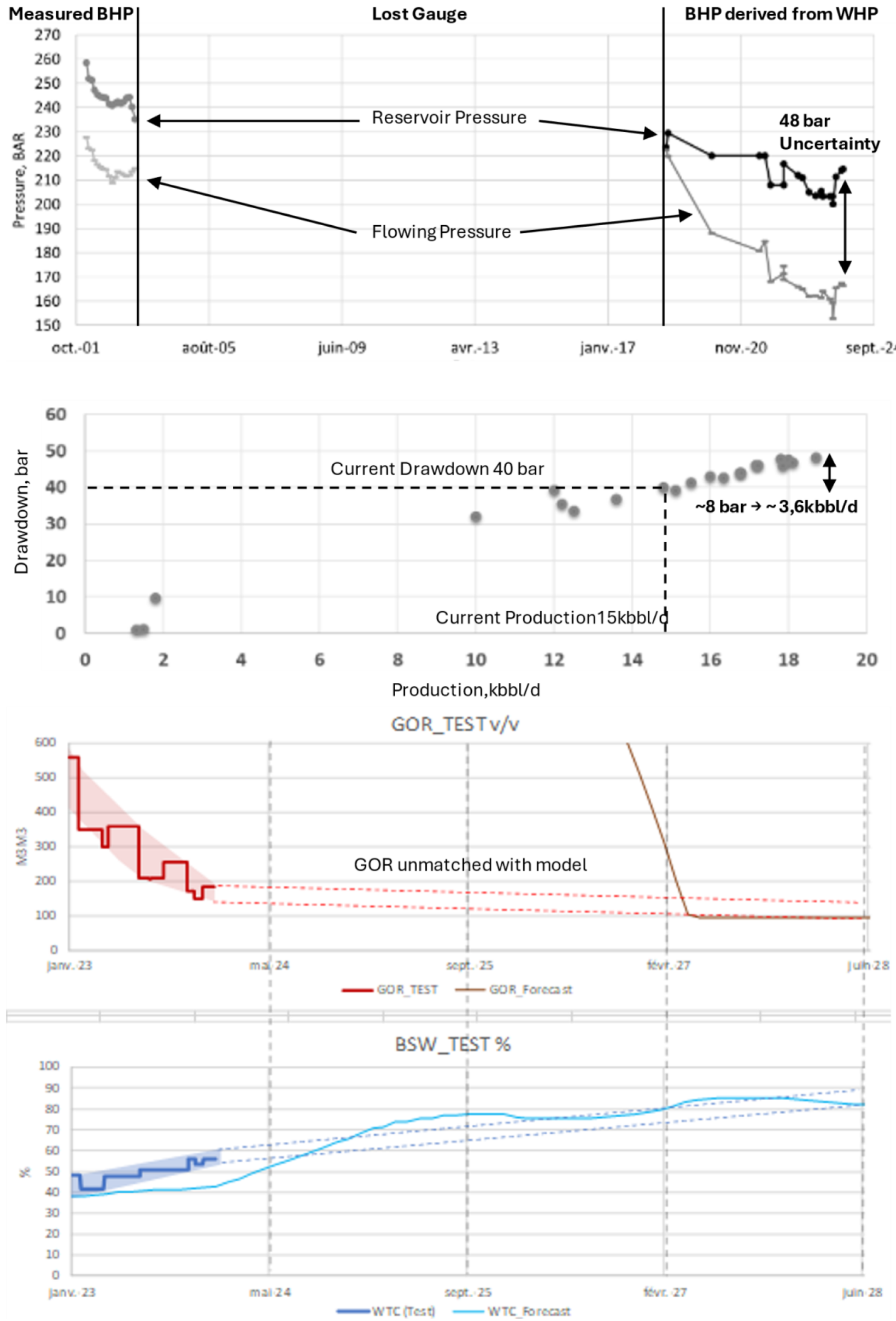

Figure 12—Value of Information case study of a deepwater field where a gauge was lost. If the gauge had been selected according to the reservoir environmental conditions, the probability of survival of the gauge would have improved significantly. The second graph shows the impact of BHP uncertainty on production. In this case, about 1MMbbl of oil in a 5-year period is lost due to the lack of pressure measurement, which requires a conservative approach in pressure/production management (7-8 drawdown margin below potential due to the risk of sand production).

In the same basin, many wells have lost downhole P/T data (up to 62% of instrumentation in the oldest field). These losses affect reservoir monitoring, production optimization, and integrity management.

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Failures stem from device aging, harsh environments, or the impact of well interventions on cables and connections. The VOI analysis highlights the importance of selecting appropriate technology. Well DW-2, for example, presents a compelling case for assuring long PDHG life-expectancy due to its current production limitations and potential upside. This particular well lacks both a PDHG and access to a multiphase flow meter (MPFM), resulting in significant uncertainty in drawdown (±5 bars) and bottom-hole pressure estimation. This uncertainty affects both production performance and well integrity, with the specific risk of screen erosion if drawdown is underestimated. Historically, DW-2 was a well candidate considered for re-stimulation, with an estimated incremental production of 1,3 MMbbl of oil in a 5-year horizon, but the plan was abandoned due to skin uncertainty derived from wellhead-based gauge data. Having a working PDHG would provide engineers with accurate pressure readings, securing productivity index (PI) and skin values. Additionally, recent riser tests suggest a significant increase in the well's BSW (from 30% to 50%), further emphasizing the need for reliable downhole data to guide operational decisions.

A different case is DW-3, an intermittent well with a history of eruptivity issues and poor pressure support from its associated injector. The well has been closed for two years due to these challenges and the failure of its PDHG. Prior to closure, DW-3 produced 3.1 kbbl/d with a liquid rate of 3.5 kbbl/d, a water cut of 20%, and a GOR of 135 vol/vol. The well showed potential during previous reopening cycles, with cumulative oil production reaching 8 MMbbl. However, without reliable downhole pressure data, its performance has been difficult to sustain. The expected oil production gain if a PDHG was still running is at least 0.7 MMbbl/year, based on the VOI calculation. A working gauge would also support water injection realignment and help monitor well integrity, particularly when it is equipped with sand control.

Conclusions and Way Forward

We have seen in the previous pages that by integrating a Business Intelligence (BI) dashboard with probabilistic analysis, for managing Permanent Downhole Gauges (PDHGs) status, marks a substantial improvement in reservoir surveillance and operational efficiency in pressure measurements. This simple, easy to deploy system addresses the limitations of traditional data collection and processing workflows by automating updates, improving data quality, and streamlining decision-making. Leveraging survival analysis techniques such as the Kaplan-Meier method, the dashboard enables robust reliability assessments of gauge technologies, facilitating proactive gauge maintenance and optimized replacement planning.

For a very limited cost, the benefits are substantial: from the data analysis perspective, the time required for monitoring surveys has been reduced from several months to just one week per country, and procurement decisions are now increasingly based on technical performance rather than cost alone. Besides, the operational benefits of using such an approach in the long term are expected to be quite significant. Enhanced drawdown management and reduced bottomhole pressure uncertainty directly contribute to production optimization, bringing additional barrels, while improved well integrity monitoring minimizes intervention needs. The case studies presented in the paper demonstrate that the absence of reliable downhole data can lead to significant production losses and operational risks, reinforcing the value of long-lasting and well-selected PDHG technologies.

Looking ahead, future developments should focus on integrating the BI dashboard with production management and predictive maintenance platforms, incorporating artificial intelligence to refine reliability models. Other statistical models can also be tested against the probabilistic approach shown here. Strengthening collaboration between reservoir, completion, and procurement teams will also be essential to ensure that technology selection is driven by performance metrics and environmental compatibility. Scaling the deployment of this tool across all assets will help standardize practices, enhance asset resilience, and maximize the value of downhole data for reservoir management. Besides, this approach could also be adapted to other instrumentation management systems, including surface or subsea equipment. Equally, in wells that are already completed, the cost of workovers to replace failed PDHGs is cost-prohibitive, and

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the presented approach only works for new drilled and completed wells. However, as shown in the VOI case studies, the benefit of having BHP and BHT could overcome the cost of intervention in some high-rate producers.

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